

Amanulla Khan (Aman)

ETL & Data Quality Test Engineer

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Goal-oriented professional, targeting **Automation, Manual, ETL and Data Warehouse Testing** to leverage expertise in SQL, Tosca, Snowflake, and Generative AI to deliver high-quality, automated, and audit-ready data validation solutions in an innovative Agile environment.

PROFILE SUMMARY



- ❖ **ETL & Data Quality Expert with 5 years** of experience **designing and executing end-to-end ETL and data warehouse testing strategies**, ensuring high-quality and reliable data pipelines across healthcare, insurance, and retail domains.
- ❖ **Automation & GenAI Innovator:** Proficient in leveraging **TOSCA, Generative AI (AVA Platform)**, and SQL-based frameworks to accelerate test design, automate data validation, and improve regression efficiency, **reducing manual effort by over 40%**.
- ❖ **Cloud & Big Data Specialist:** Skilled in validating complex datasets across **Snowflake, Azure, AWS, Aurora DB, and Databricks**, including structured, semi-structured, and large-scale delta data processing for enterprise-grade data lakes.
- ❖ **Compliance & Data Governance Advocate:** Deep understanding of **HIPAA, CMS, HITECH, and healthcare regulatory standards**, implementing test frameworks that ensure **data privacy, SCD Type 2 accuracy**, and audit-ready reporting for UAT and regulatory reviews.
- ❖ **Key Achievements & Leadership Impact:** Successfully **reduced test design time by 35%** and **defect leakage by 20% in UAT cycles** through intelligent test automation and GenAI-driven anomaly detection; recognized as **Champion of the Quarter (Q2 2024)** for delivering innovative, reusable frameworks and team mentorship.



CORE COMPETENCIES

- ETL Testing
- Automation Testing (TOSCA, GenAI)
- Healthcare & Insurance Compliance (HIPAA, CMS, HITECH)
- Test Case Design & Test Automation Frameworks
- Data Warehouse (DWH) Testing
- Data Reconciliation & Profiling
- Cloud Data Platforms (Snowflake, Azure, AWS)
- Control-M &
- UAT Support & Defect Management (Jira, q-Test)
- SQL Development & Validation
- SCD Type 2 Validation
- ETL Job Orchestration
- Metadata & Schema Validation



EDUCATION

- Bachelor of Computer Applications (BCA) - Pursuing**
University of Mysore (Distance learning)
- Associate Degree / Diploma in Engineering - 2016**
Board of Technical Education, Karnataka



SOFT SKILLS



- Stakeholder Collaboration
- Analytical Thinking
- Problem Solving
- Effective Communication
- Leadership & Team Facilitation
- Adaptability & Agile Mindset

CERTIFICATIONS

- ❖ Tricentis Tosca Deep Dive: Automating Database Test Cases
- ❖ Tricentis Tosca Fundamentals (AS1): Web Application Testing



- ❖ Tricentis Tosca Fundamentals (AS2): Centralized Test Data Management
- ❖ Tricentis Tosca Deep Dive (TDS1): Test Design and Structuring
- ❖ Tricentis Tosca Deep Dive (TDS2): Managing your stateful data
- ❖ Tricentis Data Integrity Fundamentals: End-to-End Data Warehouse Testing
- ❖ q-Test Certifications: Manager, Fundamentals, Reporting & Analytics
- ❖ Automation Tools: Excel, PDF, JSON file testing via Tosca
- ❖ Databricks Accreditation: Generative AI Fundamentals
- ❖ GenAI Certifications: Gen AI 101, 201, 301

TECHNICAL SKILLS

- ❖ **Technologies & Tools:** Power BI, Informatica PowerCenter (Workflow Manager & Monitor), Snowflake, Azure Databricks, Control-M (Batch Job Execution & Monitoring), TOSCA Data Integrity (DI), Apache Airflow, Generative AI (AVA Platform), Git, Git Bash, GitLab, GitLab CI/CD Pipelines, Boomi ETL
- ❖ **Cloud Platforms:** Snowflake, Microsoft Azure, Amazon Web Services (AWS), AWS S3 Buckets (File Ingestion & Validation).
- ❖ **Databases:** Snowflake, Aurora, Oracle, PostgreSQL, Greenplum, Teradata, Flat Files (CSV/JSON).



- ❖ **Validation Tools & Techniques:** Beyond Compare, Notepad++, Jupyter-Hub, Microsoft Excel, TOAD, Visual Studio, DBBeaver.
 - ❖ **Techniques:** Power BI Testing, Backend Testing, Schema Validation, Metadata Comparison, SCD Type 2 Validation, Data Masking, Data Reconciliation, Row- and Column-level Comparison, File-to-Table Validation.
 - ❖ **Programming Languages:** SQL.
 - ❖ **Test & Defect Management Tools:** Jira, q-Test, HP ALM, Redmine.
- ❖ **Processes & Methodologies:** Agile (Scrum/Kanban), Software Testing Life Cycle (STLC), Software Development Life Cycle (SDLC), Test Metrics & Reporting (Weekly/Monthly), Task Assignment & Team Coordination.
 - ❖ **Award:** *Champion of the Quarter – Q2 2024* - Ascendion Engineering

KEY ACHIEVEMENTS ACROSS THE CAREER

- ❖ **Recognized for automating ETL and data validation test cases using TOSCA Data Integrity (DI)**, thus **reducing manual testing effort**, expanding regression coverage and delivering **approximately 35% cost savings** across testing cycles.
- ❖ **Improved overall data quality and regulatory compliance** by applying advanced SQL-based source-to-target validations and ETL automation techniques, thereby ensuring accurate, consistent, and audit-ready data across enterprise data pipelines.
- ❖ **Led the implementation of an automated data testing framework**, which resulted in a **30% improvement in data validation accuracy** and a substantial reduction in execution time for repetitive manual validation tasks.
- ❖ **Strengthened audit readiness** by embedding reusable validation logic within **AVA Platform workflows**, produced explainable and traceable test outputs that increased stakeholder and audit confidence during UAT and compliance reviews.
- ❖ **Designed and generated synthetic healthcare datasets using GenAI-driven prompts**, enabled realistic UAT and regulatory audit simulations while safeguarding sensitive production data and improved test environment reliability.

WORK EXPERIENCE

Jul'25 – Present: Lead Engineer – Quality Engineering Ascendion Engineering, Bangalore (Remote)
Mar'23 – Jun'25: Senior Engineer – Quality Engineering Ascendion Engineering, Bangalore (Remote)
Dec'21 – May'22: Senior Engineer – QA Excel Engineers and Consultants, Pune
Sep'20 – Nov'21: Junior Executive – QA Tata Lockheed Martin Aerostructure Ltd., Hyderabad

Key Result Areas:

- ❖ **End-to-End ETL & Data Warehouse Testing:** Validated complete ETL pipelines from source systems to Snowflake Data Warehouse, reporting layers, and Databricks, performing batch, incremental, and change-data-load validations using audit columns, watermark logic, and control tables.
- ❖ **Advanced SQL & Source-to-Target Validation:** Conducted comprehensive source-to-target data verification across Snowflake, Oracle, Aurora DB, Greenplum, and flat files, including joins, aggregations, filters, lookups, and stored procedure checks.
- ❖ **ETL Automation & GenAI-Driven Testing:** Automated ETL test cases using TOSCA (DB Expert, Reconciliation, T-Box modules) and **Generative AI (AVA platform)** for test scenario creation, SQL query generation, and standardized validation steps, reducing manual effort and improving regression coverage.
- ❖ **Data Profiling, Metadata, & Compliance Testing:** Performed schema validation, data profiling, row- and column-level comparison, SCD Type 2 checks, data masking, classification logic, and role-based access control to ensure HIPAA, CMS, and HITECH compliance.
- ❖ **File-Based ETL Validation:** Tested AWS S3 inbound and outbound file processing (CSV/JSON), validating file structure, headers/trailers, control totals, rejected records, and file-to-table reconciliation.
- ❖ **Defect & Test Evidence Management:** Managed the full defect lifecycle using Jira and q-Test, prepared UAT test evidence, reconciliation reports, and validation summaries to support business sign-off and audit readiness.
- ❖ **Batch Job Orchestration & CI/CD Integration:** Triggered and monitored Control-M batch jobs, managed job dependencies, reruns, and postexecution validation, and integrated ETL automation with GitLab CI/CD pipelines for version control and deployments.
- ❖ **Regression & Enhancement Support:** Executed automated weekly regression suites, quickly regenerated updated test cases and validation queries for business logic or stored procedure changes, reducing defect leakage and test cycle time.
- ❖ **Agile Collaboration & Metrics Reporting:** Actively participated in Agile ceremonies (daily stand-ups, sprint planning, retrospectives), tracked testing tasks, and presented weekly/monthly QA metrics including execution status, defect trends, and automation coverage.
- ❖ **Team Leadership & Mentorship:** Acted as QA point of contact, coordinated with developers, data engineers, analysts, and stakeholders, and mentored junior team members on SQL validation, ETL testing best practices, TOSCA automation, CI/CD processes, and healthcare data compliance.

PROJECTS

Client: Centene Corporation – Healthcare & Insurance Domain | Location: United States
Tools/Technologies: Snowflake, Aurora DB, MongoDB, TOSCA, AVA Platform, DBeaver, Control-M, Jira, GitHub
Project Overview: Worked on validating large-scale healthcare and insurance data pipelines supporting eligibility, claims processing, provider matching, and member enrolment across multiple states for a Fortune 500 healthcare organization.

- Key Contributions & Achievements:**
- ❖ Automated **40% of test cases** using **AVA platform** with context-aware prompts for validating stored procedures, masking rules, and classification logic for In-sprint Stories.
 - ❖ Designed prompt-based SCD Type 2 validations, improved edge-case coverage and reducing human errors.
 - ❖ Built synthetic healthcare datasets using GenAI to simulate UAT and regulatory audit scenarios while protecting sensitive data.
 - ❖ Developed robust test flows to ensure data integrity across insurance claims, eligibility, denial, and payment pipelines.
 - ❖ Embedded validation logic within AVA workflow, generated explainable and audit-ready test outputs.
 - ❖ Achieved **35% faster test design and 20% reduction in defect leakage** in UAT cycles through intelligent anomaly detection and prompt engineering.



PERSONAL DETAILS

Date of Birth: 12nd September 1997 | **Languages Known:** English, Hindi, Kannada, Marathi | **Address:** Bangalore, India